

Week 5 Feasibility Memo: AI-Assisted Emergency Department Triage

Verdict

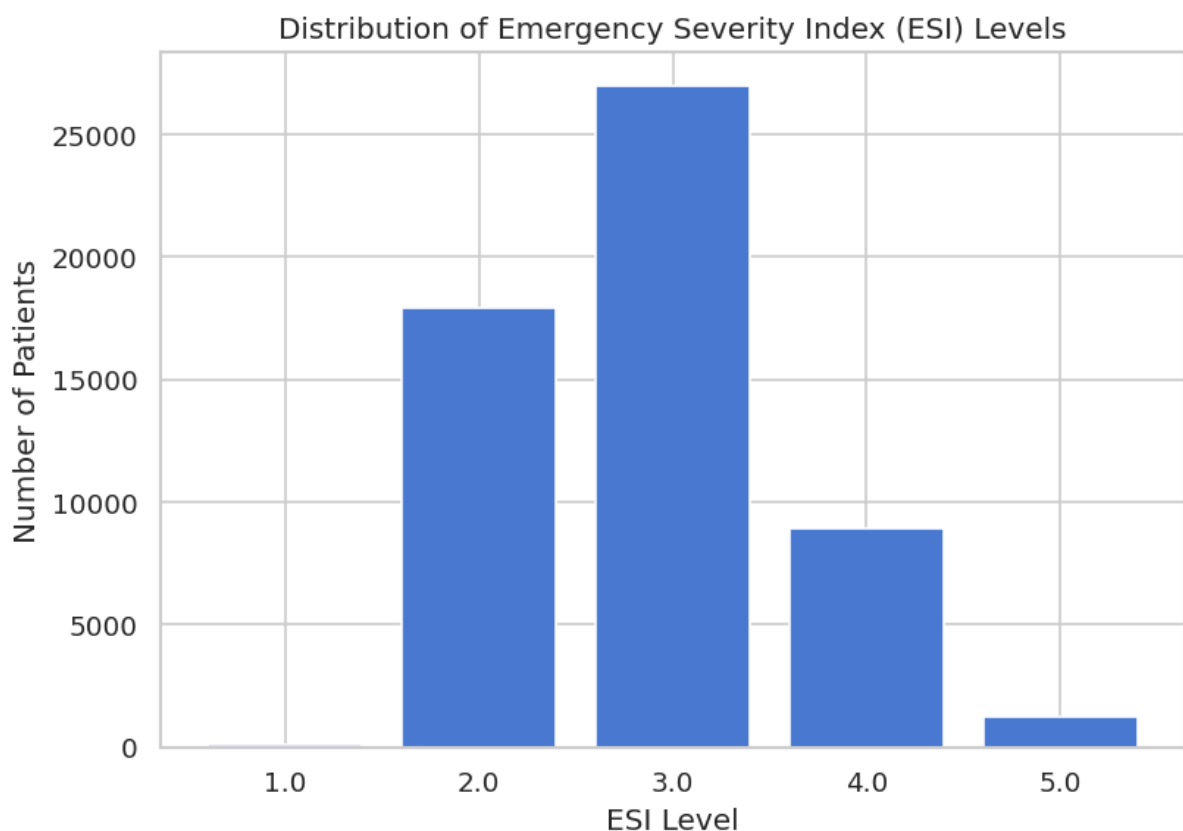
The Yale EMLLC Emergency Department Triage & Admission Prediction dataset is suitable for developing an AI-assisted triage model, provided that future work addresses class imbalance, external validation, and fairness before clinical deployment.

Dataset Summary

This exploratory analysis evaluated the Yale EMLLC Emergency Department Triage & Admission Prediction dataset to determine its suitability for supporting an AI-assisted emergency department triage system. The dataset contains a large number of emergency department encounters and includes demographic information, vital signs collected during triage, historical healthcare utilisation, and chief complaint indicators. These variables reflect information that is routinely available shortly after patient arrival, making them appropriate inputs for a triage-support model.

The exploratory analysis examined dataset completeness, data types, statistical outliers, variable distributions, and relationships between numerical variables and the Emergency Severity Index (ESI). Overall, the released dataset demonstrates a high level of completeness and contains clinically meaningful information that supports further predictive modelling.

Figure 1 (Emergency Severity Index distribution) illustrates that patient presentations are not evenly distributed across triage categories, with comparatively fewer patients assigned to the highest-acuity levels. This imbalance should be considered during model development because it may affect predictive performance for critically ill patients.



Top Three Data Quality Concerns

1. Evidence of Prior Preprocessing

The released dataset contains minimal observable missing data across the analysed variables. This suggests that substantial preprocessing, such as imputation, exclusion of incomplete records, or other data quality procedures, was performed before publication. Although this improves readiness for analysis, it limits the ability to evaluate the original pattern of missingness and understand how incomplete records were managed.

2. Class Imbalance

The distribution of Emergency Severity Index (ESI) levels is uneven, with relatively few patients classified in the highest-acuity groups. Models trained without appropriate techniques may therefore favour more common triage categories and perform less effectively when identifying critically ill patients.

3. Limited Generalisability

The dataset originates from a single healthcare system. Differences in patient populations, clinical workflows, and local triage practices may limit the direct transferability of any resulting model to other hospitals without additional validation.

Top Three Reasons to Proceed

1. Clinically Relevant Variables

The dataset contains a broad range of variables routinely collected during emergency department triage, including age, vital signs, previous healthcare utilisation, and chief complaint indicators. These features are clinically relevant and available early enough to support real-time decision-making.

2. High Data Completeness

The exploratory analysis identified very little observable missing data in the released dataset, allowing the analysis to proceed without extensive data cleaning and reducing the likelihood of missing-value-related modelling challenges.

3. Meaningful Physiological Relationships

Exploration of heart rate, blood pressure, respiratory rate, oxygen saturation, and temperature demonstrated clinically plausible variation across Emergency Severity Index categories. Correlation analysis also identified meaningful relationships between several physiological variables and triage acuity, supporting their inclusion in future predictive models.

Caveats

The exploratory analysis provides evidence that the dataset is appropriate for model development, but it does not demonstrate that an AI model would be safe or effective in clinical practice. Statistical associations identified during exploration should not be interpreted as causal relationships.

Future work should include feature engineering, model comparison, fairness assessment across demographic groups, calibration analysis, and external validation using data from additional healthcare organisations. Given the observed imbalance in Emergency Severity Index categories, appropriate strategies such as class weighting or resampling should also be considered to ensure reliable performance across all triage levels.

Conclusion

The Yale EMLC Emergency Department Triage & Admission Prediction dataset provides a strong foundation for AI-assisted emergency department triage research. It combines clinically relevant variables with high overall data completeness and demonstrates meaningful relationships between

physiological measurements and triage acuity. Although limitations relating to preprocessing, class imbalance, and generalisability remain, these challenges are common in healthcare machine learning and can be addressed during subsequent stages of model development and evaluation. Overall, the exploratory analysis supports proceeding to predictive modelling while maintaining appropriate attention to fairness, validation, and clinical governance.

Github link: [carisurg-portfolio/docs/Week 5 at main · dontayrosemain7-git/carisurg-portfolio](https://github.com/dontayrosemain7-git/carisurg-portfolio/docs/Week%205%20at%20main)